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1. INTRODUCTION:

In modern automobiles, security has become an essential feature to prevent unauthorized access and enhance user convenience. Traditional key-based systems are being replaced with smart, wireless control systems for improved safety. This project focuses on developing a Bluetooth-based automobile security system using Arduino technology. The system allows users to control basic car functions such as locking, unlocking, and ignition through a smartphone.

Communication between the smartphone and the automobile is established via the HC-05 Bluetooth module. Simulates a car security system controlled via Bluetooth. Instead of using a physical key, the system allows the car to lock/unlock and simulate ignition using commands sent from a smartphone over Bluetooth. Demonstrates secure wireless communication, micro-controller control, and actuator simulation inside a model car or a hardware platform.

The Arduino microcontroller processes the received commands and controls actuators like LEDs, buzzer, relay, and DC motor. Security is ensured by authenticating the user with a password before granting access to car operations. The system provides feedback through LEDs and buzzer signals, along with confirmation messages sent back to the smartphone. This project demonstrates a low-cost, wireless, and user-friendly alternative to traditional vehicle security systems. Overall, it highlights the application of embedded systems and wireless communication in developing smart automobile solutions.

2. COMPONENTS USED:

The components used in the making of this project is given as follows

- Arduino UNO
- Bluetooth Module: HC-05
- Relay
- LED
- Buzzer
- DC motor
- Bluetooth controller mobile app
- Jumper Wire
- Breadboard

2.1 ARDUINO UNO:



Fig. 1.1: Arduino UNO Board

Arduino is an open-source electronics platform based on easy-to-use hardware and software. It allows users to build digital devices and interactive objects. Arduino boards are equipped with a micro-controller that can be programmed. The most common board is the Arduino Uno, using the ATmega328P chip. You write code in the Arduino IDE, which uses a simplified version of C/C+. It can read inputs (like sensors) and control outputs (like motors or LEDs). Arduino is great for beginners, hobbyists, and prototyping.

It supports both USB and external power sources. The Crystal oscillator has a frequency of 16MHz, which makes the Arduino UNO a powerful board. Voltage Regulator - The voltage regulator converts the input voltage to 5V. GND - Ground pins. The ground pin acts as a pin with zero voltage. Vin - It is the input voltage.

2.2 BLUETOOTH MODULE- HC-05:

The HC-05 is a popular Bluetooth module used for wireless serial communication. It operates using the Bluetooth 2.0 protocol (classic Bluetooth). It can be configured to work as either a master or slave device. The module communicates with micro-controllers using UART (TX/Rx).

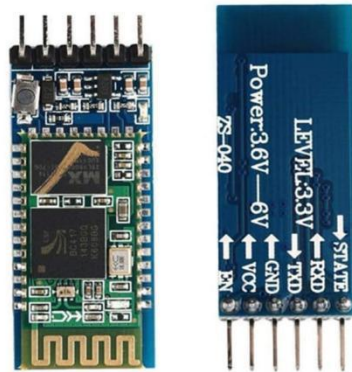


Fig .1.2: Bluetooth Module -HC 05

Its default baud rate is 9600 bps, but it can be changed via AT commands. It operates on 3.3V logic, but can be powered with 5V through its onboard regulator. It has onboard status LED's to show connection . HC-05 has a range of about 10 meters in open space. It's commonly used with Arduino for wireless data transfer. The HC-05 Bluetooth module is a widely used wireless communication device designed for serial communication with micro-controller like Arduino. It operates on the 2.4 GHz frequency and supports Bluetooth 2.0+EDR. The module communicates via UART allowing it to either connect to other devices or be connected to. With a range of around 10 meters, it's commonly used in home automation, robotics, and wireless data transmission projects.

2.3 RELAY:

A relay is an electromagnetically switch used to control high-voltage or high-current devices. It works by using a small electrical signal to open or close a larger electrical circuit. A relay typically has a coil, common (COM), normally open (NO), and normally closed (NC) contacts. When the coil is energized, it magnetically pulls a switch, changing the connection. Relays are used to control devices like lights, fans, motors, and appliances.



Fig. 1.3: Relay Module

They are isolated, meaning the control circuit is separate from the load circuit. Standard relays operate at 5V or 12V for control, but can switch 220V AC loads. In Arduino projects, relay modules simplify interfacing with micro-controllers. Solid-state relays (SSR) offer faster, quieter switching without mechanical parts. Relays are essential in automation, home control, and safety systems. A relay is an electrically operated switch used to control a high-voltage or high-current circuit with a low-power signal, like from a micro-controller. It consists of an electromagnet that opens or closes contacts when energized.

2.4 LED:



Fig. 1.4: LED

An LED is a semiconductor device that emits light when current flows through it. It stands for Light Emitting Diode. LED's have two terminals: the anode (+) and the cathode (-). Current flows from anode to cathode, and the LED lights up. They are energy-efficient and generate less heat than traditional bulbs. LEDs come in various colors (red, green, blue, white, etc.). They are commonly used in indicators, displays, lighting, and electronics. LED's require a resistor in series to prevent damage from excess current.

In Arduino projects, they are often used to show status or output. LED's are long-lasting, reliable, and widely used in modern technology. LED or Light Emitting Diode is a light source device. It releases light when current passes through it. The LED concept is based on a semiconductor device where holes and electrons recombine to produce energy in the form of photons. The need for light-emitting materials comes from the need for the wavelength of radiation. LED's are used in different applications, such as decorative lights, remote control circuits, etc.

2.5 BUZZER:

A buzzer is an audio signaling device that produces sound when powered. It converts electrical energy into sound energy. Buzzers are used for alerts, alarms, notifications, and sound feedback. There are two main types: active (has internal oscillator) and passive (needs external signal). An active buzzer produces sound with just a DC voltage.



Fig. 1.5: Buzzer

A passive buzzer requires a PWM signal to generate tones. Buzzers operate typically at 5V, making them easy to use with Arduino. They can generate simple beeps, tones, or melodies. Commonly used in timers, security systems, and electronic projects. A buzzer is an audio signaling device that produces a sound when electrical voltage is applied. It is commonly used for alerts, alarms, notifications, and user feedback in electronic devices. Buzzers can be active

(produces sound when powered) or passive (requires an external signal to produce sound). They typically operate on low voltages (e.g., 3V to 12V) and are easy to interface with micro-controller like Arduino.

2.6 DC MOTOR:

A DC motor is an electric motor that runs on direct current (DC) electricity. It converts electrical energy into mechanical rotational motion. It has two main parts: a rotor (armature) and a stator (magnetic field). When current flows, the motor shaft rotates, allowing movement in devices. The direction of rotation can be changed by reversing the polarity of the power supply. Speed can be controlled using PWM (Pulse Width Modulation) signals. DC motors come in various sizes and power ratings. Common types include brushed and brush-less DC motors. Widely used in robots, fans, toys, and electronic tools.



Fig.1.6: DC Motor

A DC motor is an electric motor that converts direct current electrical energy into mechanical rotation.

It operates on the principle that a current-carrying conductor in a magnetic field experiences a force, causing the motor shaft to rotate. They typically require a power supply ranging from a few volts to several dozen volts, depending on the motor size. Common types include brushed and brushless DC motors.

2.7 BREAD BOARD:

A breadboard is a tool used to build and test electronic circuits without soldering. It has a grid of holes (sockets) for inserting electronic components and wires. The internal connections allow components to be electrically connected by simply plugging them in. The horizontal rows in the middle are usually for placing components. The vertical columns on the sides are typically used for power rails (VCC and GND). It is reusable, making it ideal for prototyping and experimenting. No permanent connections are made, so designs can be changed easily. Breadboards come in various sizes: mini, half-size, and full-size.

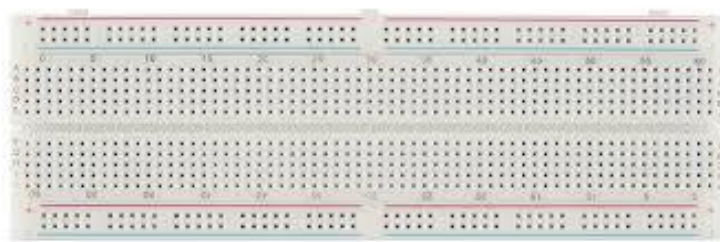


Fig .1.7: Full- size Bread Board

Often used in Arduino and other electronics projects for quick testing. A breadboard is a widely used tool to design and test circuit. You do not need to solder wires and components to make a circuit while using a bread board. It is easier to mount components & reuse them. Since, components are not soldered you can change your circuit design at any point without any hassle.



It consist of an array of conductive metal clips encased in a box made of white ABS plastic, where each clip is insulated with another clips.

2.8 JUMPER WIRES:

A jumper wire is an electrical wire used to connect components on a breadboard or circuit. They come with connector pins on both ends, usually male or female. Types include male-to-male, male-to-female, and female-to-female. Jumper wires are essential for prototyping and testing circuits without soldering. They are typically used with breadboards, Arduino, and other development boards. Made with flexible insulated wire and standard 2.54 mm (0.1 inch) connectors. Available in various colors to help organize connections. Allow for quick and temporary circuit modifications.



Fig.1.8: Jumper wires-

It is useful for connecting sensors, LED's, modules, and power rails. A jump wire is an electrical wire, or group of them in a cable, with a connector or pin at each end which is normally used to

interconnect the components of a breadboard or other prototype or test circuit, internally or with other equipment or components, without soldering. Individual jump wires are fitted by inserting their "end connectors" into the slots provided in a breadboard, the header connector of a circuit board, or a piece of test equipment. Generally, jumpers are tiny metal connectors used to close or open a circuit part. They have two or more connection points, which regulate an electrical circuit board.

3. CIRCUIT IMAGE:

The circuit image of the Bluetooth-Based Car Security System is given below.

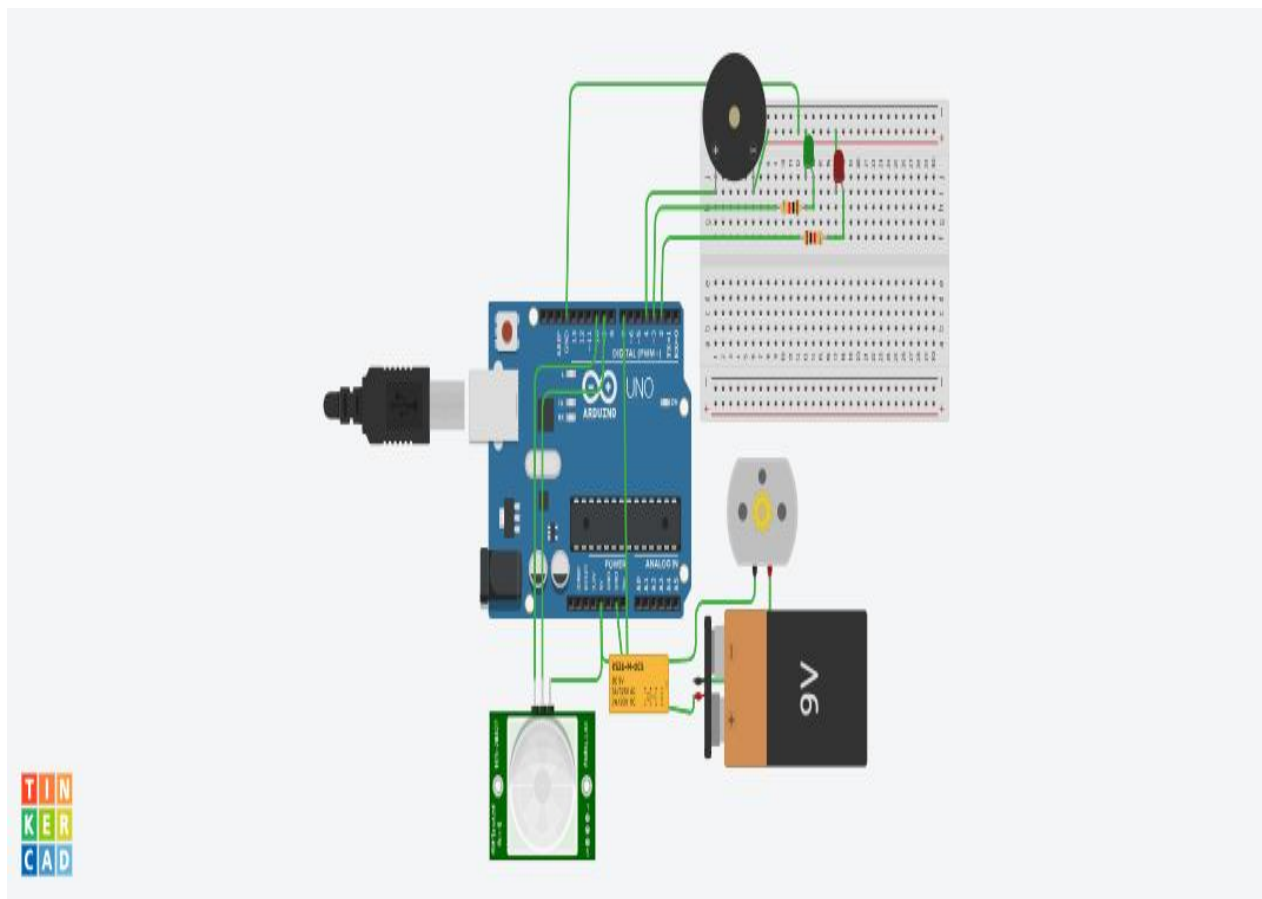


Fig. 1.9: Circuit diagram of Bluetooth-Based Car Security System

4. ALGORITHM:

The algorithm of this is given in the following steps

Step 1: Start System

Step 1.1: Power ON Arduino, Bluetooth module (HC-05), and other components.

Step 1.2: Initialize Serial communication and set all pins (LED s, buzzer, relay,motor) as OUTPUT.

Step 2: Bluetooth Pairing

Step 2.1: Smartphone pairs with HC-05 (default PIN: 1234).

Step 2.2: Wait for connection establishment.

Step 3: Authentication

Step 3.1: Arduino waits for password command in format PWD:1234.

Step 3.2: If password correct → set authenticated = true.

Step 3.3: Else → send “WRONG PASSWORD” message and request again.

Step 4: Command Reception

Step 4.1: Continuously listen for incoming commands from smartphone.

Step 4.2: Read the received string and trim extra spaces.

Step 5: Command Processing

Step 5.1: If authenticated:

Step 5.1.1: LOCK → Turn ON Red LED, Turn OFF Green LED, beep buzzer, send “CAR LOCKED”.

Step 5.1.2: UNLOCK → Turn ON Green LED, Turn OFF Red LED, beep buzzer, send “CAR UNLOCKED”.

Step 5.1.3: START → Activate relay → run DC motor (simulate engine start), send “ENGINE STARTED”.

Step 5.1.4: STOP → Deactivate relay → stop DC motor, send “ENGINE STOPPED”.

Step 5.1.5: Any other command → send “INVALID COMMAND”.

Step 5.2: If not authenticated → request password again.

Step 6: Actuator Feedback

Step 6.1: LEDs indicate car status (Red = Locked, Green = Unlocked).

Step 6.2: Buzzer beeps for confirmation of LOCK/UNLOCK.

Step 6.3: Relay + Motor simulate ignition on START/STOP commands.

Step 7: Loop

Step 7.1: Repeat steps 4–6 continuously while system is powered ON.

Step 8: End

Step 8.1: System remains idle until new commands are received or power is turned OFF.

5. WORKING:

1. Power ON & Initialization

- When the system is powered ON, the Arduino initializes all components (Bluetooth, LED's, relay, motor, buzzer).
- The Bluetooth module (HC-05) becomes discover-able for pairing with a smartphone.

2. Bluetooth Connection & Authentication.

- The smartphone connects to the HC-05 module.
- Before using commands, the user must send a password (e.g., PWD:1234).
- If the password matches, access is granted; otherwise, the system denies commands.

3. Command Sending from Smartphone

- The user sends commands through a Bluetooth control app (e.g., Bluetooth Terminal).
- Supported commands are:
LOCK → Lock car.
UNLOCK → Unlock car.
START → Start ignition.
STOP → Stop ignition.

4. Command Reception & Processing

- Arduino receives the command via Bluetooth serial communication.
- It checks which command has arrived and executes the respective operation.

5. Operation Execution

- LOCK: Red LED glows, buzzer beeps shortly, indicating the car is locked.
- UNLOCK: Green LED glows, buzzer beeps, indicating the car is unlocked.
- START: Relay activates the DC motor to simulate engine start.
- STOP: Relay turns OFF, stopping the DC motor (engine stopped).

6. Feedback to User

- The Arduino sends back a response message via Bluetooth (e.g., “CAR LOCKED”, “ENGINE STARTED”).
- LED's and buzzer provide physical confirmation of the action.

7. Security Features

- Only paired and authenticated devices can send commands.
- Unauthorized users cannot control the car without password access.

8. Continuous Monitoring

- The system runs in a loop, continuously listening for new commands until power is switched OFF.

6. OUTPUT:

The following output demonstrates the functioning of the Bluetooth-based Automobile Security System.

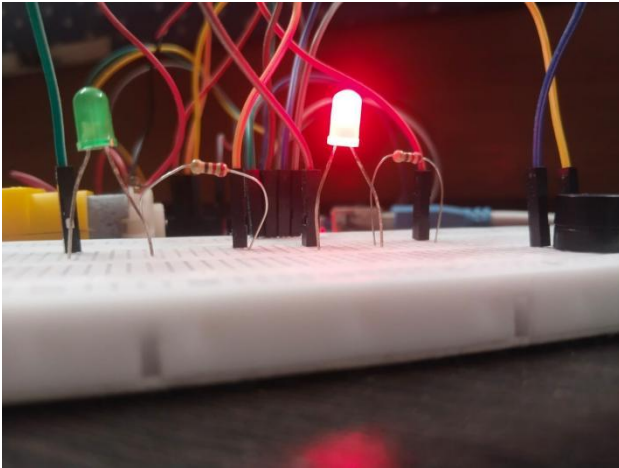


Fig. 1.10: Red LED turned ON after the LOCK command

- In the first image, the red LED is glowing, which indicates that the car door is locked.
- Along with the red LED, the buzzer also gives a short beep to confirm the locking action.
- This state ensures that unauthorized access to the car is prevented, just like in a real automobile security system.

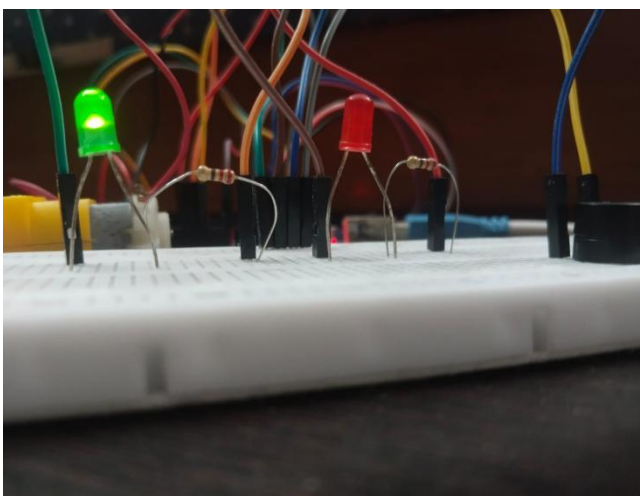


Fig. 1.11: Green LED turned ON after the UNLOCK command

- In the second image, the green LED is glowing, which represents that the car door is unlocked.
- When unlocked, the user can access the car or proceed to the next command, such as starting the ignition.
- The switching between red and green LED's provides clear visual feedback of the car's security status.

CONCLUSION:

The project successfully implements a low-cost and reliable car security system using Bluetooth technology. Only authorized users with paired device can access the vehicle, ensuring safety against theft. The system is a simple, user-friendly and eliminates the need for traditional keys. The Bluetooth-Based Car Security System successfully demonstrates the integration of wireless communication and embedded systems to enhance vehicle safety and control. By using Bluetooth technology, the system allows secure, short-range wireless access to car locking mechanisms, replacing traditional keys with smartphone-based control. This not only adds convenience for the user but also reduces the vulnerability of mechanical locks to common theft techniques.

Through the use of a micro-controller (e.g., Arduino), a Bluetooth module (e.g., HC-05), and actuators like relays, the system effectively controls vehicle access based on user authentication. It is low-cost, easily implementable, and highly customization, making it ideal for both personal and commercial vehicle security applications.